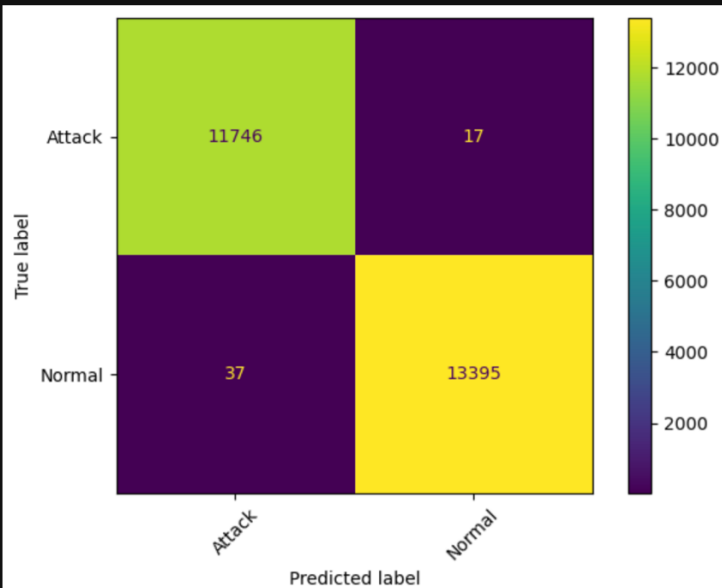


Nick Leung EE446 Assignment 2 Discussion

Accuracy 0.9978567361831665

Classification Report

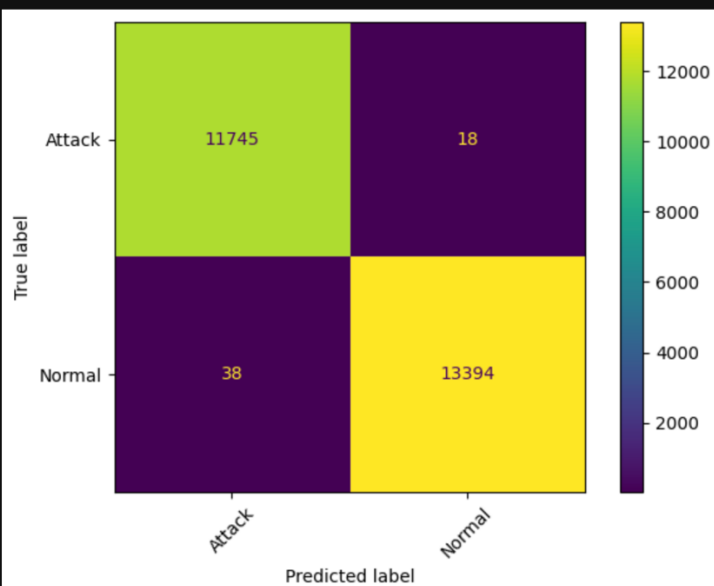
	precision	recall	f1-score	support
0	1.00	1.00	1.00	11763
1	1.00	1.00	1.00	13432
accuracy			1.00	25195
macro avg	1.00	1.00	1.00	25195
weighted avg	1.00	1.00	1.00	25195



3.

Classification Report

	precision	recall	f1-score	support
0	1.00	1.00	1.00	11763
1	1.00	1.00	1.00	13432
accuracy			1.00	25195
macro avg	1.00	1.00	1.00	25195
weighted avg	1.00	1.00	1.00	25195



4.

5c. When I tried to deploy the dynamic model, I got an error saying the tensor failed to invoke. In setup, I saw an error message saying hybrid models aren't supported by one of the libraries. With some debug prints in setup() using the dynamic model, I got these errors:

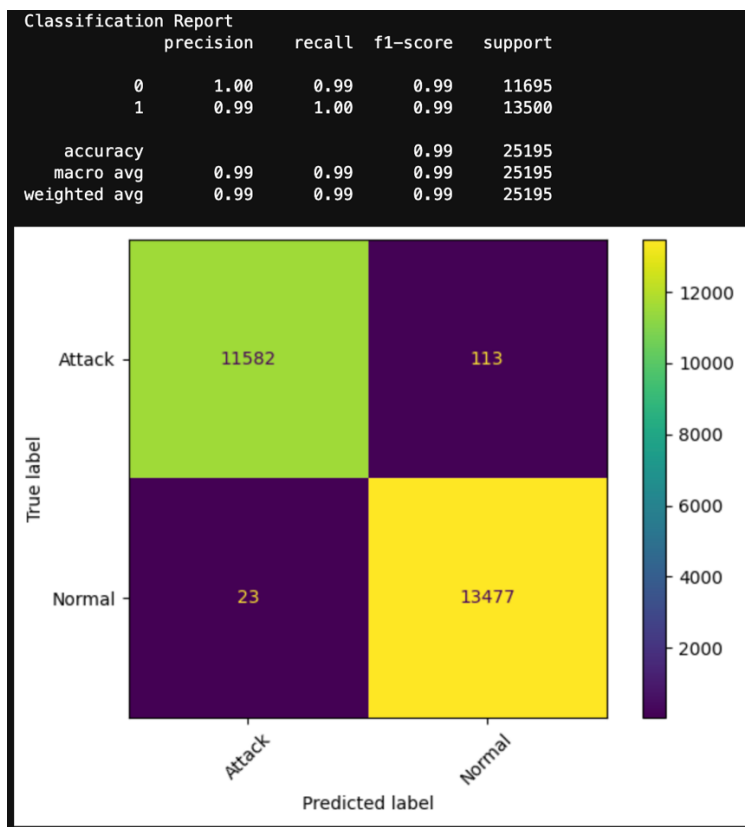
```
SERIAL BEGIN
Model loaded
/Users/nick/Documents/Arduino/libraries/Arduino_TensorFlowLite/src/tensorflow/lite/micro/kernels/cmsis-nn/fully_connected.cpp Hybrid models are not supported on TFLite Micro.
Node FULLY_CONNECTED (number 0f) failed to prepare with status 1
Failed to allocate tensors!
MicroAllocator: Model allocation started before finishing previously allocated model
Failed starting model allocation.

Failed to invoke!
Invoke() called after initialization failed
```

To solve this issue, I quantized my base model to int8 and got the following results:

```
Sample #: 1 | Predicted class: Normal | Actual class: Normal
Sample #: 2 | Predicted class: Normal | Actual class: Normal
Sample #: 3 | Predicted class: Attack | Actual class: Attack
Sample #: 4 | Predicted class: Normal | Actual class: Attack
Sample #: 5 | Predicted class: Normal | Actual class: Normal
```

Int8 evaluation metrics:



5d.

I used the same int8 model to get these results:

Sample #:	1	Predicted class:	Attack	Actual class:	Attack
Sample #:	2	Predicted class:	Normal	Actual class:	Normal
Sample #:	3	Predicted class:	Attack	Actual class:	Attack
Sample #:	4	Predicted class:	Normal	Actual class:	Normal
Sample #:	5	Predicted class:	Normal	Actual class:	Normal
Sample #:	6	Predicted class:	Normal	Actual class:	Normal
Sample #:	7	Predicted class:	Normal	Actual class:	Normal
Sample #:	8	Predicted class:	Normal	Actual class:	Attack
Sample #:	9	Predicted class:	Normal	Actual class:	Attack
Sample #:	10	Predicted class:	Attack	Actual class:	Attack